Proposal selection and continuation-stable diagnostics on finite graphs

Jeremy H. Carroll

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Abstract

We separate the generation of candidate edge fields from a decision rule based on independently computed diagnostics. On a finite connected graph, distance to the image of the incidence map measures failure of exactness. A declared face map supplies a closure residual. A lexicographic comparison of diagnostics, subject to a ratio condition, defines one proposal-selection policy. Equal diagnostic traces have equal decisions by construction, but equal current diagnostics need not remain equal after further proposals. For a declared family of transformations, we characterize the largest equivalence relation inside diagnostic equality that is preserved by every finite continuation. We give explicit cycle examples distinguishing local exactness from global exactness, and one-step decisions from continuation-stable state compression. Executable paths additionally require agreement of adjacent endpoint states. The accompanying exact rational checks and Lean results are scoped to their stated finite and abstract objects.

1. Setting and purpose

Let $P = (V, E)$ be a finite directed graph whose underlying undirected graph is connected. A field $\omega \in \mathbb{R}^E$ assigns a real value to each directed edge. Reversing the orientation of an edge reverses the sign used in a path sum. The incidence map is

$$(Bf)_{ij} = f(j) - f(i).$$

A proposal is a specified map T taking a field to a candidate field. The selection rule studied here evaluates the candidate field rather than the proposal's name or description. This separation is an interface definition; it does not establish that an arbitrary software system follows that definition. The graph and convex-optimization foundations are standard [1,2].

2. Quotient residuals and cycle consistency

Define

$$\Phi(\omega) = \min_f \|\omega - Bf\|_1.$$

The minimum is attained: the residual space is affine and closed, and its intersection with a sufficiently large norm ball is compact. Potentials need not be unique; fixing one vertex removes the additive constant but does not in general remove all minimizer nonuniqueness.

For a signed graph cycle C, write $\Delta_C(\omega) = \sum_{e \in C} \epsilon_e \omega_e$, where the sign records agreement with the traversal orientation.

Theorem 2.1. The following are equivalent:

1. $\Phi(\omega) = 0$.
2. $\omega = Bf$ for some potential f.
3. $\Delta_C(\omega) = 0$ for every cycle C.

Proof. Attainment makes the first two equivalent. If $\omega = Bf$, summing around a cycle telescopes to zero. Conversely choose a root and define $f(v)$ as the signed sum along a path from that root to v. Two such paths differ by a cycle-space element; the zero-cycle condition makes the sum independent of the path. Applying this to the endpoints of an edge gives $Bf = \omega$. Checking a cycle basis suffices by linearity.

On a tree every field is exact. On a cyclically oriented triangle,

$$\Phi(\omega) = |\omega_1 + \omega_2 + \omega_3|.$$

Every residual has this same signed sum, giving the lower bound by the triangle inequality. A potential can set two residual coordinates to zero, attaining it.

For rational data, an exact primal–dual witness consists of a potential f, residual $h = \omega - Bf$ and vector φ such that

$$B^T\varphi = 0, \qquad \|\varphi\|_\infty \le 1, \qquad \varphi^T\omega = \|h\|_1.$$

For any other potential a, dual feasibility gives $\varphi^T\omega = \varphi^T(\omega - Ba) \le \|\omega - Ba\|_1$. Thus the equality certifies optimality without trusting the solver that supplied the witness. Existence of a witness for every arbitrary numerical input is a separate solver question.

3. Closure residuals and ratio conventions

When faces are specified, let D be their signed edge-to-face map and require $DB = 0$. For any minimizing residual $h = \omega - Bf^*$,

$$\Gamma(\omega) = \|Dh\|_1 = \|D\omega\|_1 \le \|D\|_{1\to 1}\Phi(\omega).$$

The equality follows from $DB = 0$; the inequality is the induced norm bound. Closure is therefore independent of the chosen minimizer. Without faces one may use a zero-dimensional target and $D = 0$, giving $\Gamma = 0$ by explicit convention; no additional face information is thereby recovered.

For $\Phi > 0$ the exact ratio is Γ/Φ. A numerical policy can instead define, for $\epsilon \ge 0$,

$$R_\epsilon = \begin{cases} \Gamma/\Phi, & \Phi > \epsilon, \\ 0, & \Phi \le \epsilon \text{ and } \Gamma \le \epsilon, \\ +\infty, & \Phi \le \epsilon < \Gamma. \end{cases}$$

The infinity value marks rejection under the policy; it is not an infinite geometric curvature. With $\Phi = 1/4000$, $\Gamma = 1/2000$ and $\epsilon = 1/1000$, the policy ratio is zero while the exact ratio is two. Passing a ceiling on R_ϵ consequently does not establish the same ceiling on the exact ratio.

4. Decisions based on diagnostic traces

Let $d(\omega) = (\Phi(\omega), \Gamma(\omega), X(\omega))$, where X is a declared auxiliary function. If it depends on a selected minimizing residual, the selection convention is part of the definition of d. If no such function has been independently specified, setting $X = 0$ supplies no extra information.

One exact policy accepts $\omega \to \omega'$ when $R_\epsilon(\omega') \le R_{\max}$ is finite and the triple $d(\omega')$ is strictly lexicographically smaller than $d(\omega)$. This means a decrease of the first coordinate, or a tie there and a decrease of the second, or ties in both and a decrease of the third.

Proposition 4.1. Two proposals with the same before/after diagnostics and the same ratio-test value receive the same decision under this policy.

Proof. The decision is the stated function of those values. The proposal label is not an argument of that function.

This is a property of the specified interface, not a claim that identical labels have identical effects. An implementation must compute or verify the diagnostic values from the candidate fields. Caller-supplied numbers alone do not certify the connection to those fields. Strict real-valued descent also need not terminate: the sequence $1, 1/2, 1/3, \ldots$ strictly decreases indefinitely.

5. Examples of information lost by a diagnostic

Local exactness does not imply global exactness

On a directed triangle take $\omega = (1, 0, 0)$. Restrict it to each of the two-edge path subgraphs. Each path is a tree, so each restriction has zero quotient residual. The restrictions agree on shared edges and assemble into the given global field, whose residual is one. Compatible local edge values can therefore form a global field without making that field exact. This example distinguishes gluing of data from the additional cycle condition for gluing potentials.

A periodic proposal need not be selected

Cyclically shifting the three coordinates of $(1, 0, 0)$ produces a period-three orbit. Every field in that orbit has cycle sum one and the same quotient residual. With $D = 0$ and $X = 0$, none of those steps meets the strict selection rule. Noncommuting transformations likewise need not be distinguished by a diagnostic: their different outputs may be mapped to the same diagnostic value.

Equal current diagnostics can separate after a continuation

Let $x = (1, 0, 0)$ and $y = (-1, 0, 0)$ on the triangle, again with $D = 0$ and $X = 0$. Both have $d = (1, 0, 0)$. For the translation $T(w) = w + (1, 0, 0)$,

$$\Phi(Tx) = 2, \qquad \Phi(Ty) = 0.$$

Thus equality of current diagnostics does not permit identification of the two states if subsequent translations must be represented correctly.

6. Continuation-stable equivalence

Let X be any state set, $d : X \to Y$ an observation map and $\mathcal{T}$ a declared family of total maps $X \to X$. Write $\mathcal{T}^*$ for finite compositions, including the identity. Define

$$x \sim y \quad \iff \quad d(Wx) = d(Wy) \text{ for every } W \in \mathcal{T}^*.$$

Theorem 6.1. This is the largest equivalence relation contained in $\ker d$ that is preserved by each map in $\mathcal{T}$.

Proof. Reflexivity, symmetry and transitivity hold pointwise after every continuation. The identity continuation gives $\sim \subseteq \ker d$. If $x \sim y$ and $T \in \mathcal{T}$, then every $W \circ T$ is another finite continuation, so $d(WTx) = d(WTy)$; hence $Tx \sim Ty$.

Let $R \subseteq \ker d$ be an equivalence relation preserved by all admitted maps. Induction on word length gives $Wx \, R \, Wy$ whenever $x \, R \, y$. Containment in $\ker d$ gives $d(Wx) = d(Wy)$ for each word, so $R \subseteq \sim$. This establishes maximality.

The quotient supports well-defined maps $[T]([x]) = [Tx]$ and $\bar{d}([x]) = d(x)$. It is the coarsest state quotient preserving all the specified finite observations and transformations. If $\ker d$ is already preserved by every admitted map, it equals $\sim$. At the other extreme, if d is nonconstant and all total maps are admitted, $\sim$ is equality: any two distinct states can be sent to two states distinguished by d.

These statements are relative to the specified family. They do not provide an algorithm for computing this relation on an arbitrary infinite state space.

7. Finite vector-valued fields

The residual definition extends to a fixed norm g on $\mathbb{R}^k$:

$$\Phi_g(\omega) = \min_{f:V\to\mathbb{R}^k} \sum_{e\in E} g(\omega_e - (Bf)_e).$$

Finite-dimensionality again gives attainment. Coordinatewise application of the path-sum argument shows that this residual is zero exactly when the vector-valued field is exact. Choosing $g = \|\cdot\|_1$ separates the problem into scalar channels; choosing $g = \|\cdot\|_2$ couples them within each edge. Neither choice is selected by the scalar quotient construction alone. The bundled reference solver is scalar; this formula is a mathematical extension, not a claim that the solver implements every vector-valued norm problem.

Matrix-based proposals need no privileged role in selection. For example, if $A^T = -A$ and $B^T = -B$, then $[A, B]^T = -[A, B]$. Commutators therefore stay in the space of real skew-symmetric matrices. A particular generating family need not generate that entire Lie algebra, and this closure observation imposes no residual decrease on a proposed field.

8. Paths, implementation and formal scope

A list of individually accepted steps need not be an executable trajectory. Adjacent steps must also satisfy equality of the preceding target state and following source state. Diagnostic equality at the boundary does not imply state equality, as the translation example demonstrates.

code/mod_operator_proposal_core.py implements exact rational witness checks, the diagnostic comparison and endpoint checks. Its general graph solver uses a two-phase simplex over rational numbers and independently checks the resulting primal–dual equality. The existence of the mathematical optimum does not imply that every possible input representation or solver implementation is correct; the witness checker supplies a separate condition on each returned result.

lean4/Proposal.lean verifies integer triangle identities, preservation of a finite path predicate under label erasure, and the continuation-equivalence properties for arbitrary state and observation types. Its finite path example uses natural-number before/after values; it is not a mechanization of every real-valued graph diagnostic or numerical solver. No external formal theorem names or absent source files are needed for the results cited here.

9. Discussion

Diagnostic equality has several distinct uses: defining equal one-step decisions, assembling compatible edge data, and identifying states under future transformations. The first does not imply the other two. A selected proposal satisfies the stated comparison policy; further properties such as feasibility, constraint preservation and convergence require their own evidence. The continuation relation gives an exact criterion for state identification relative to a declared transformation family, without turning that criterion into an unrestricted compression claim.

References

1. A. Hatcher, *Algebraic Topology*, Cambridge University Press, 2002, Sections 2.1 and 3.1. Author-hosted book.
2. S. Boyd and L. Vandenberghe, *Convex Optimization*, Cambridge University Press, 2004, Chapters 4–5. Author-hosted book.